

From Code to Copilots

Measuring the rise of AI/ML open-source activity on GitHub, 2021–2026 · 85,339,283 events · 273.5 GB · PySpark on GCP Dataproc

THE QUESTION

How has open-source activity around AI/ML repositories changed compared with general software-development repositories from 2021 to 2026?

“Everyone knows AI is booming” is an assumption, not a measurement. We measured it from the raw public record — every public GitHub event, one day per quarter, 23 quarters.

1 · PROFILING

Characterize 273.5 GB at the command line — never loading it into memory.

552 hourly .json.gz · 35 GB gzip
273,494,892,306 bytes uncompressed
85,339,282 events (zcat | wc -l)
→ 55× the 5 GB floor, 1.7× the 50 M floor

zcat | wc -l took 38m37s as one stream, 13m26s across 10 cores — identical bytes. Streaming reads only what it needs: head -1 returns in 13 ms regardless of pile size.

2 · BREAKING

Make the in-memory tools fail on purpose, reproducibly.

EXCEL — 1,048,576-row hard cap. Fed 1.6 M rows (8 hours of ONE day) it loaded 1,048,575 and silently dropped 35% of the data. Our full dataset is 81× the cap.

PANDAS — read_json on ONE hour (1.0 GB of JSON) peaked at 14.4 GB resident: 14× amplification. One day under an 8 GB cap died with MemoryError.

3 · SCALING

Same spark_job.py, same 273.5 GB, read straight from gs://. Only the cluster changes.

cluster	job time	speedup
single-node	43m37s	1.00×
2 workers	17m28s	2.50×
4 workers	10m21s	4.21×

All three runs produced byte-identical CSVs (md5 match): adding workers changed the wait, not the answer. The like-for-like step 2→4 is 1.69× on 2× cores = 84% efficiency, capped because gzip is non-splittable — 552 files are 552 fixed tasks.

273.5 GB in → 43 KB out → this poster.

THE ANSWER

2.35×

AI/ML activity growth
21Q1 → 26Q1

1.28×

general activity growth
same window

0.94%

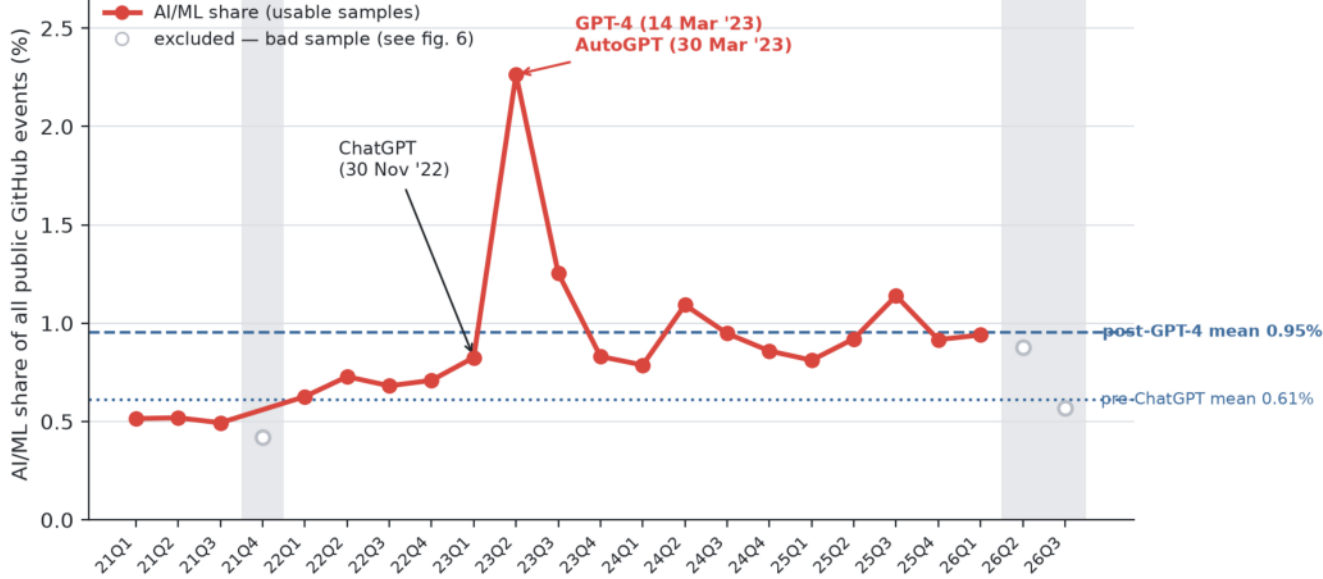
AI/ML share of ALL
public GitHub events

4.03×

more likely to be STARRED
than a general repo

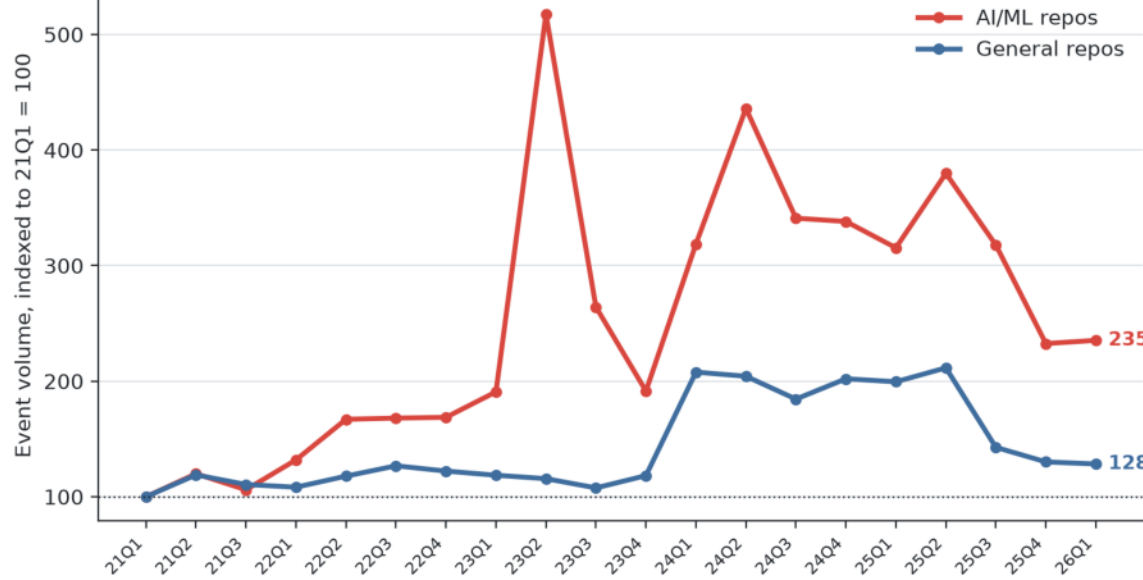
The AI boom shows up on GitHub as an attention boom first, and a code boom second.

AI/ML repositories captured a lasting, but smaller-than-the-hype, share of GitHub



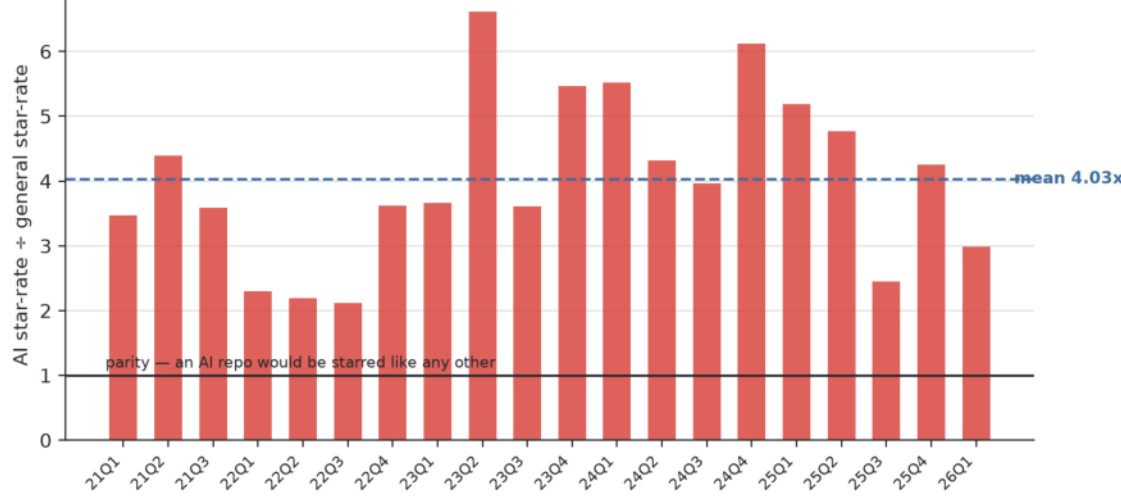
Share peaks at 2.264% in 23Q2 — 4.4× the 21Q1 baseline — in the quarter of GPT-4 (14 Mar '23) and AutoGPT (30 Mar '23), one quarter after ChatGPT. And it was new people: distinct actors in AI repos went 3,734 → 19,977 in that single quarter (5.4×). Most left by 23Q3, but a 1.56× structural lift remains — mean share 0.61% before ChatGPT vs 0.954% after GPT-4.

AI/ML activity grew 2.35x while general activity grew 1.28x



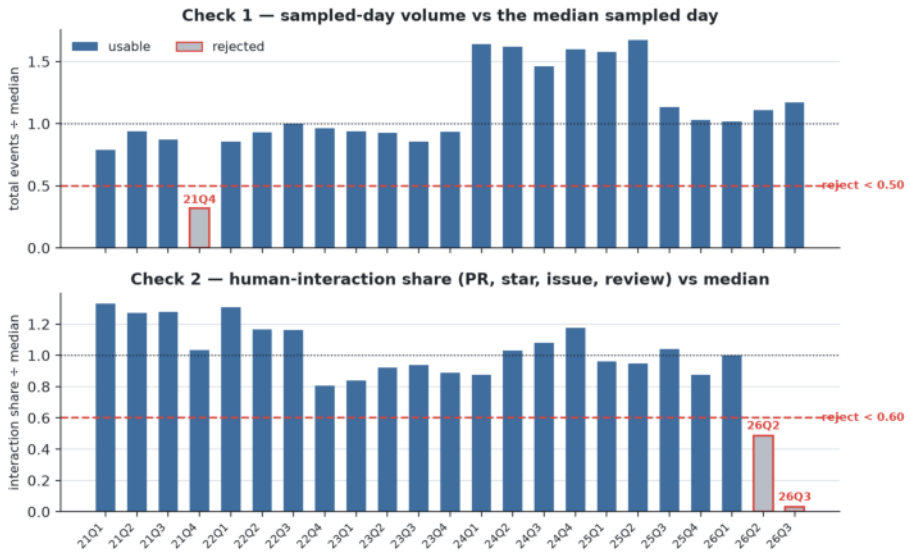
AI/ML compounded at 18.7%/yr against 5.1%/yr for general software — 3.7× faster. But it started from 0.5% of the platform, so even after the largest hype cycle in software history, AI/ML repos remain under 1% of all public GitHub events. Both of those are true, and the poster says both.

Every single usable quarter: AI/ML repos are starred at several times the general rate



THE DURABLE FINDING — stars and forks are consumption signals; pushes are production. Pooled across usable quarters, AI/ML repos are starred 4.4× and forked 3.1× more intensively than general repos while being pushed to LESS (43.5% vs 59.2% of their events). The ratio never drops below parity in any of the 20 usable quarters, ranging 2.12×–6.61×. The AI corner of GitHub is somewhere people come to look at and take from, not to build in.

Three of 23 sampled days are unusable — and they fail for two different reasons



DATA QUALITY — every point rests on ONE sampled day, so a single bad day becomes a fake trend. Two automated checks reject 3 of 23 days before any finding is computed: 21Q4 holds only 32% of a median day's events (missing hourly files), while 26Q2 and 26Q3 kept normal push volume but lost pull requests, stars and comments upstream (general PullRequestEvent 304,163 → 10,806). The –68.65% sitting in monthly_growth.csv is that artifact, not a collapse in AI activity.

WHAT WOULD BREAK THIS ANSWER

The AI/ML classifier is a keyword list frozen in 2021 — llm, gpt, pytorch, langchain, transformer, huggingface, openai, neural. It cannot match repos named claude, mcp, cursor, copilot, gemini, vllm or ollama, because those tools did not exist when the list was fixed. AI activity in 2024–26 is therefore systematically undercounted, and 2.35× is a floor rather than a ceiling. It also means part of the 23Q2 spike is a naming event: that was the quarter everything got named *gpt*.

Also: repo name ≠ repo content; bots inflate pushes far more than stars, which strengthens the attention finding; and 23 sampled days stand in for five years.